

Automated SOC Analyst Playbook Using Splunk, n8n, and AI-Powered Threat Analysis

Author

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Project Type

Security Automation & Incident Triage

Technologies

Splunk Enterprise, n8n, AI Agent, OpenAI/Groq LLM, REST API Integration

Executive Summary

This project demonstrates the development of an automated SOC analyst workflow capable of collecting security telemetry from Splunk Enterprise, processing event data through an AI-powered analysis engine, and generating automated threat assessments.

The objective was to reduce analyst workload by automating the initial triage phase of security investigations. Using n8n as the orchestration platform, the workflow retrieves live security logs from Splunk via API, forwards the data to an AI agent for contextual analysis, and returns a summarized security assessment highlighting suspicious activity and operational findings.

The project successfully demonstrates how Security Operations Centers can leverage automation and artificial intelligence to improve incident response speed, reduce alert fatigue, and accelerate security investigations.

Project Objectives

The primary goals of this project were:

- Automate Splunk log collection.
- Reduce manual analyst review time.
- Enrich raw telemetry with AI-driven analysis.
- Demonstrate automated security triage.
- Build a repeatable SOC investigation workflow.

- Simulate next-generation AI-assisted SOC operations.
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Security Architecture

User Query



n8n Workflow Trigger



Splunk REST API



Log Collection



AI Agent



Large Language Model



Threat Analysis



Investigation Summary



SOC Analyst Review

Environment Overview

Component	Function
Splunk Enterprise	Security Log Repository
n8n	Workflow Automation
AI Agent	Security Context Processor
OpenAI/Groq LLM	Threat Analysis Engine
Windows Systems	Telemetry Source
Sysmon	Endpoint Visibility

Security Use Case

Security analysts frequently spend significant time reviewing large volumes of log data to determine whether activity represents a genuine threat.

This project automates the first stage of the investigation process by:

1. Retrieving security logs automatically.
2. Processing event context.
3. Identifying suspicious behaviors.
4. Summarizing findings.
5. Presenting investigation-ready output.

The workflow functions as a virtual Tier 1 SOC analyst capable of performing initial triage operations.

Workflow Breakdown

Phase 1: Security Telemetry Collection

The workflow begins when a security investigation request is initiated.

An HTTP Request node establishes communication with the Splunk Management API.

API Endpoint

<https://192.168.10.10:8089/services/search/v2/jobs/export>

Query Configuration

Search Mode:

search index=*

Execution Mode:

oneshot

Output Format:

json

Purpose

Retrieve recent security events from Splunk without requiring manual analyst interaction.

Phase 2: Data Processing

The collected JSON data is normalized and forwarded to the AI Agent node.

The AI Agent serves as the investigation coordinator responsible for:

- Understanding event context.
 - Identifying suspicious activity.
 - Prioritizing findings.
 - Generating investigation summaries.
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Phase 3: AI-Assisted Security Analysis

The AI Agent forwards event telemetry to a Large Language Model.

Model Used

llama-3.3-70b-versatile

Analysis Functions

- Process execution review
- Network activity analysis
- User behavior inspection

- Event correlation
- Security summarization

The AI engine transforms raw telemetry into human-readable security findings.

Investigation Results

Process Execution Analysis

The workflow identified a PostgreSQL process running under the SYSTEM security context.

Event Details

Process Name:

postgres.exe

User Context:

SYSTEM

Process ID:

13424

Security Assessment

The AI correctly recognized the process as a legitimate PostgreSQL database service.

The workflow successfully differentiated normal administrative activity from suspicious behavior.

This demonstrates contextual awareness rather than simple keyword matching.

Network Activity Analysis

The AI engine detected DHCP-related network activity.

Observed Activity

Network Interface:

em0

DHCP Server:

10.128.120.158

Security Assessment

The activity was classified as expected infrastructure communication.

No evidence of:

- Command and Control traffic
- Data exfiltration
- Lateral movement
- Unauthorized network scanning

was identified.

AI Investigation Output

The generated output included:

- Process descriptions
- User context analysis
- Network event interpretation
- Security relevance assessment
- Investigation summaries

The workflow converted raw machine telemetry into analyst-readable intelligence.

Security Benefits

Reduced Investigation Time

Traditional SOC Workflow:

Analyst → Search Logs → Review Events → Correlate Findings

Estimated Time:

15–30 minutes

Automated Workflow:

Workflow → AI Analysis → Summary Generated

Estimated Time:

Less than 1 minute

Alert Fatigue Reduction

The AI layer filters and summarizes large event datasets before they reach analysts.

Benefits include:

- Faster triage
 - Reduced manual review
 - Improved analyst efficiency
 - Better prioritization
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Detection Engineering Value

The project demonstrates several core SOC capabilities:

Capability	Implemented
Log Collection	Yes
API Integration	Yes
Security Automation	Yes
Event Correlation	Yes
AI Analysis	Yes
Automated Triage	Yes
Investigation Support	Yes

Security Risks & Considerations

While AI provides substantial operational benefits, security teams must validate findings before taking action.

Potential risks include:

- False positives
- Hallucinated conclusions
- Missing context
- Overreliance on automation

Human analyst validation remains essential.

Future Enhancements

Recommended improvements include:

1. MITRE ATT&CK mapping.
 2. IOC extraction.
 3. Threat scoring.
 4. Automated ticket generation.
 5. Slack or Microsoft Teams notifications.
 6. Splunk notable event integration.
 7. Automated threat hunting workflows.
 8. Multi-source telemetry ingestion.
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Lessons Learned

This project demonstrated how AI and automation can enhance SOC operations by accelerating the investigation lifecycle.

The workflow successfully:

- Retrieved live telemetry from Splunk.
- Processed security data automatically.
- Generated contextual threat assessments.
- Reduced manual analyst effort.
- Improved visibility into security events.

The exercise highlights the growing importance of AI-assisted security operations within modern Security Operations Centers.

Conclusion

The Automated SOC Analyst Playbook successfully demonstrated the integration of Splunk Enterprise, n8n workflow automation, and AI-powered threat analysis.

The solution collected security telemetry, performed automated event review, generated contextual security assessments, and delivered analyst-ready findings within seconds.

This project showcases practical skills in SOC automation, SIEM integration, API engineering, AI-assisted threat analysis, incident triage, and security orchestration—capabilities increasingly required in modern SOC and Blue Team environments.

The workflow represents a foundational SOAR-style solution capable of enhancing analyst productivity and improving operational efficiency within security teams.